

Federated Learning for Financial Fraud Detection: Privacy, Heterogeneity, and Operational Trade-offs

A research synthesis for cross-institution model training without centralized raw transaction data

Literature-based research synthesis

Abstract

Financial fraud detection benefits from large and diverse transaction histories, but institutions often cannot pool raw customer data because of confidentiality, security, legal, and competitive constraints. Federated learning (FL) offers a different training topology: participating institutions compute model updates locally and a coordinator aggregates those updates into a shared model. This arrangement can reduce the need to centralize raw records, yet it does not automatically solve privacy or governance. Fraud detection also presents unusually difficult learning conditions, including extreme class imbalance, non-identically distributed client data, concept drift, asymmetric error costs, and adversarial behavior. This paper synthesizes foundational federated optimization research and representative studies of federated credit-card fraud detection. It proposes an architecture and evaluation protocol for cross-institution fraud models, emphasizing precision-recall metrics, cost-sensitive thresholds, robust aggregation, secure update handling, and local validation. The central argument is that federated learning should be viewed as a coordination mechanism rather than a privacy guarantee. Practical deployments require defense-in-depth: secure aggregation, access governance, update validation, heterogeneity-aware optimization, imbalance handling, drift monitoring, and institution-level rollback procedures.

Keywords: federated learning, fraud detection, financial services, privacy, class imbalance, non-IID data

1. Introduction

Fraud detection is a data-intensive classification problem with a structural tension. Models improve when they observe diverse legitimate and fraudulent behavior, yet the most valuable transaction data is distributed across banks, card issuers, processors, merchants, and payment platforms. Centralizing those records can create security, confidentiality, regulatory, and operational risk. Federated learning changes the data-flow assumption: rather than moving raw examples to a shared repository, participants train locally and send model information for aggregation.

The canonical Federated Averaging algorithm introduced by McMahan et al. (2017) demonstrated that local optimization followed by periodic model averaging can reduce communication relative to synchronizing every gradient step. In finance, Yang et al. (2019) applied the idea to credit-card fraud detection so that institutions could contribute to a shared detector without directly exchanging their transaction datasets. Subsequent work has explored data balancing, adaptive aggregation, and privacy-enhancing mechanisms.

However, “data stays local” is not equivalent to “privacy is solved.” Model updates can still create leakage risks, malicious clients can submit poisoned updates, and a global model may perform poorly for institutions whose data distribution differs from the federation average. Fraud detection intensifies these problems because fraud is rare, attacker behavior changes, institutions serve different customers and geographies, and the cost of false positives is operational rather than purely statistical. This paper asks how FL can be designed and evaluated as a realistic cross-institution fraud-detection system.

2. Fraud Detection as a Federated Learning Problem

In a simplified horizontal FL setting, each institution k holds examples drawn from its own distribution and trains the same model architecture. During round t , the coordinator distributes global parameters w_t . Each selected institution performs local optimization and returns updated parameters $w_{(t+1)}^k$ or an equivalent update.

Federated Averaging computes a weighted aggregate, commonly proportional to local sample counts: $w_{(t+1)} = \text{sum}_k (n_k / N) * w_{(t+1)}^k$. The equation is simple; the assumptions behind it are not.

Fraud data are typically non-IID across institutions. One bank may have a large e-commerce portfolio, another may emphasize corporate cards, and a third may operate in a region with different merchant patterns. Even the definition and investigation outcome of “fraud” can differ. Statistical heterogeneity means that local gradients may point in conflicting directions. Systems heterogeneity adds another problem: participants can have different compute budgets, network conditions, batch sizes, and training schedules.

The class distribution is also highly skewed. Niu et al. (2019), using a widely studied public credit-card dataset, evaluated supervised and unsupervised models under a setting in which fraudulent transactions were only a tiny fraction of observations. In a federation, imbalance can exist both globally and locally: one institution may observe many confirmed cases while another sees almost none. As a result, sample-count weighting can favor institutions with abundant legitimate transactions rather than those contributing the most informative fraud examples.

3. Review Approach

This paper uses a focused narrative review of foundational federated optimization, privacy-preserving aggregation, and representative fraud-detection studies. It includes FedAvg as the baseline coordination method, FedProx and FedNova as examples of algorithms designed for heterogeneous federated settings, secure aggregation as a protocol-level protection, and studies that apply federated learning directly to card-fraud detection. The objective is to identify design trade-offs rather than to rank papers whose datasets and evaluation protocols differ.

4. Cross-Institution Reference Architecture

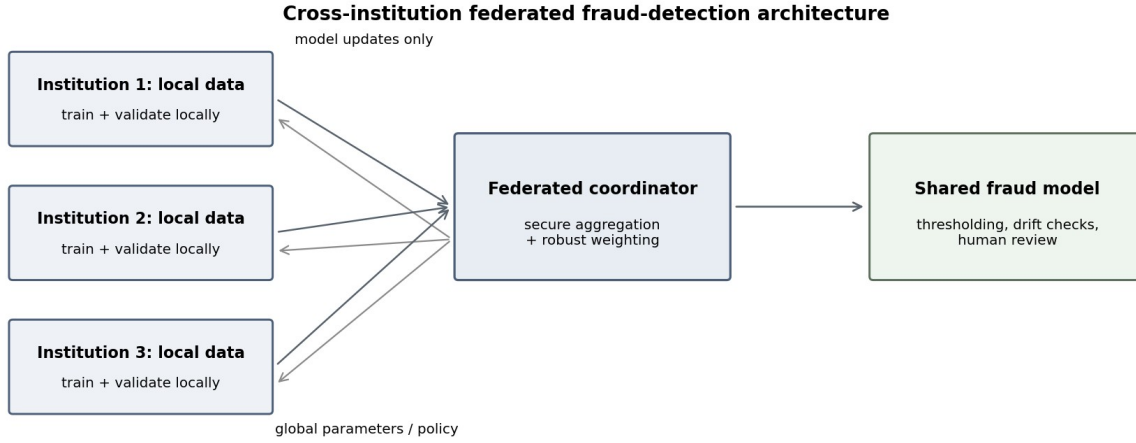


Figure 1. Cross-institution architecture with local training, a federated coordinator, and local operational decision layers.

4.1 Local data and feature contracts

Federated training requires more standardization than the phrase “keep data local” suggests. Participants need compatible feature semantics, label definitions, model interfaces, and update schedules. A transaction feature called merchant risk score must mean the same thing across participants or be transformed into a shared representation. If institutions use different fraud-confirmation windows, the global model may learn inconsistent labels. Governance therefore begins with a feature and label contract rather than with the aggregation server.

Raw features do not have to be identical in every design. Vertical or representation-based federation can support different feature sets, but complexity rises quickly. For a first cross-bank fraud deployment, horizontal federation around a common feature schema is operationally simpler and easier to audit.

4.2 Local training and imbalance handling

Each institution should evaluate its local class distribution before every training cycle. Common responses to imbalance include class-weighted losses, threshold tuning, under-sampling, over-sampling, or synthetic minority generation. Yang et al. (2019) combined federated learning with oversampling in their FFD framework. More recent work by Abdul Salam et al. (2024) compared multiple resampling approaches and classifiers within a federated credit-card fraud setting, reinforcing the point that class balancing and federated optimization should be evaluated jointly rather than as independent modules.

For operational fraud systems, threshold selection is as important as model probability quality. A threshold that maximizes F1 may create an unacceptable number of manual reviews. Each institution may therefore need a local decision threshold even when the scoring model is shared. This preserves a common representation of fraud risk while allowing operations teams to reflect local staffing capacity, customer tolerance, and loss severity.

4.3 Aggregation under statistical heterogeneity

FedAvg can work well when local objectives are sufficiently aligned, but heterogeneity can destabilize training. FedProx adds a proximal term that discourages local models from drifting too far from the global parameters and was designed specifically for statistical and systems heterogeneity (Li et al., 2020). FedNova analyzes objective inconsistency created when clients perform different amounts of local work and normalizes updates to reduce bias (Wang et al., 2020). Adaptive federated optimizers such as federated versions of Adam and Yogi provide another route for coping with difficult optimization behavior (Reddi et al., 2021).

No aggregation method is universally best for fraud detection. The choice should be benchmarked under realistic client partitions rather than random IID splits. A random split can make federated learning appear nearly identical to centralized learning while hiding the actual deployment problem. Evaluation should deliberately model institutions with different fraud rates, customer mixes, transaction volumes, and feature drift.

4.4 Privacy and security controls

Federated learning reduces the need to pool raw data, but client updates remain sensitive. Secure aggregation protocols are designed so that the coordinator learns an aggregate of client contributions without seeing each individual update in the clear. Bonawitz et al. (2017) developed a practical secure aggregation protocol for privacy-preserving machine learning in which aggregation remains possible despite client dropouts. This protects update visibility but does not by itself prevent a malicious participant from submitting a harmful update.

Defense-in-depth can include authenticated participants, secure channels, hardware-backed key storage, update clipping, anomaly detection on model deltas, robust aggregation, differential privacy, audit logging, and staged rollout. Each mechanism has a cost. Differential privacy can reduce leakage risk but may reduce model utility, especially when the positive class is already rare. Robust aggregation can limit outliers but may suppress legitimate updates from an institution whose distribution is genuinely different. Security controls therefore need threat-model-specific evaluation.

5. Evaluation: Accuracy Is Not Enough

Fraud detection is an imbalanced decision problem, so overall accuracy can be misleading. A classifier can achieve very high accuracy by predicting most transactions as legitimate. Evaluation should emphasize area under the precision-recall curve (AUPRC), fraud-class recall, precision at an operational alert budget, false-positive rate, expected monetary cost, calibration, and time-to-detection. Metrics should be reported both globally and per institution.

Dimension	Example measures	Why it matters
Detection quality	AUPRC; recall; precision; recall at fixed alert rate	Captures rare-event performance
Business cost	Expected fraud loss + review cost + customer friction	Connects statistical errors to operating consequences
Fairness across clients	Per-institution metrics; worst-client performance	Prevents a strong average from hiding weak participants
Federation efficiency	Rounds to convergence; bytes exchanged; client runtime	Measures practical training overhead
Privacy/security	Update exposure; attack success; leakage tests	Tests whether coordination creates new information risk
Robustness	Performance under drift, dropouts, poisoned clients	Assesses behavior outside the clean benchmark

Table 1. Recommended evaluation dimensions for federated fraud detection.

A strong experiment should also compare at least four baselines: local-only models, a hypothetical centralized model where legally and technically possible for research, basic FedAvg, and the proposed federated method. Local-only performance reveals whether collaboration actually helps. The centralized model estimates the statistical price paid for decentralization. FedAvg reveals whether additional complexity is justified.

6. Major Failure Modes and Mitigations

Failure mode	Operational mechanism	Representative mitigations
Non-IID drift	Institutions learn conflicting local patterns	FedProx/FedNova-style methods, clustering, personalization
Local class scarcity	Some clients see almost no confirmed fraud	Class weighting, resampling, shared representation, longer windows
False-positive overload	High recall creates too many investigations	Local thresholding, cost-sensitive tuning, alert budgets
Update leakage	Model updates reveal information about training data	Secure aggregation, differential privacy, minimization
Poisoned updates	Compromised client changes the global model	Authentication, clipping, robust aggregation, anomaly checks
Concept drift	Fraud tactics and legitimate behavior change	Time-sliced validation, drift monitors, rapid retraining
Weak-client harm	Global model improves average but hurts one bank	Per-client metrics, personalization, rollback and opt-out

Table 2. Failure modes specific to a cross-institution fraud federation.

7. Governance and Deployment Model

The federation coordinator should be governed like shared financial infrastructure rather than like an ordinary model-training server. Participant onboarding needs identity, technical conformance, and legal agreements defining permitted use of updates. Model versions should be signed and reproducible. Training rounds should produce auditable records of participating institutions, software versions, validation results, aggregation configuration, and rollback points, while avoiding logs that recreate sensitive data.

Deployment should separate training from decision authority. A shared model can produce a risk score, but individual institutions may retain the final transaction threshold, intervention type, and case-management process. This separation is useful because operational costs differ. It also allows a bank to temporarily fall back to a previous model if the new global version degrades local validation.

A staged lifecycle is preferable: offline retrospective validation, shadow scoring, limited production rollout, and only then broad adoption. Each stage should include adversarial testing and drift analysis. Federated learning can reduce data-centralization risk, but poor governance can simply replace one concentration of risk with another at the model coordinator.

8. Research Agenda

The most important research gap is evaluation under realistic cross-bank heterogeneity. Many federated studies simulate clients by randomly partitioning a single dataset, which is convenient but may underestimate distribution shift. Future benchmarks should preserve institution-level temporal and behavioral differences, include delayed fraud labels, and evaluate performance under changing attack patterns.

A second direction is personalized federation. A single global fraud detector may be too coarse when institutions operate different products. A shared backbone with local heads, cluster-specific models, or lightweight adapters may capture common fraud structure while preserving local specialization. The challenge is to determine when personalization reflects legitimate heterogeneity and when it merely overfits local noise.

Third, privacy and robustness need joint analysis. Stronger privacy mechanisms can make it harder to inspect suspicious updates, while robust aggregation may require signals about update geometry. Designing protocols that preserve confidentiality while still supporting attack detection is a central systems problem for high-stakes federations.

9. Limitations

This paper is a literature-based research synthesis and does not present new transaction-level experiments. Published fraud studies use different data sources, client simulations, imbalance treatments, model architectures, and metrics, so numerical results are not directly comparable. The governance recommendations are intended as design principles and must be mapped to the legal and operational requirements of the institutions involved.

10. Conclusion

Federated learning is attractive for fraud detection because it enables collaborative model training without requiring all raw transaction data to be centralized. Its value is therefore architectural, but its success is statistical and operational. Non-IID client data, rare fraud labels, asymmetric costs, model-update privacy, malicious participants, and concept drift can all undermine a naive federation. The strongest design treats FL as one layer in a controlled system: standardized features and labels, heterogeneity-aware optimization, imbalance-aware learning, secure aggregation, per-institution evaluation, cost-sensitive thresholds, and explicit rollback. Under those conditions, federated learning can become a credible way to share predictive strength without sharing the underlying transaction records themselves.

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